

GUIDELINE FOR THE MANAGEMENT OF DIABETIC KETOACIDOSIS (DKA) IN ADULTS

First Name: Surname:
DOB: Hosp No:
NHS No:

IMPORTANT: For children under 18 years of age please use the appropriate DKA pathway.

STEP 1

Initial Assessment

- ABCDE
- GCS
- NEWS/MEWS
- Capillary Blood Glucose (CBG)
- Ketones (blood)
- Weight

Obtain IV access and take bloods:

FBC ☐ U&E ☐ Lab Glucose ☐ VENOUS Blood Gas ☐

If unable to obtain IV access, request senior doctor / critical care support immediately.

← Initial here to confirm that you have assessed this patient correctly.

STEP 2

Confirm the Diagnosis

Criteria	Record Value	Compare Value	Criteria met?
Capillary Blood Glucose		Above 11mmol/L	YES ☐
Capillary Ketones		Above 3mmol/L	YES ☐
Venous pH -or- Bicarbonate (HCO₃)		pH less than 7.3 or HCO ₃ less than 15mmol/L	YES ☐

← Initial here if all 3 boxes are ticked (unless all 3 criteria are met this is not DKA).

Important: If not DKA, the patient may need a Variable Rate Intravenous Insulin Infusion (VRIII) – consult a senior doctor for advice.

Important: When diagnosis confirmed, immediately inform critical care outreach team and diabetes team.

STEP 3

Initial Fluid Resuscitation

Is systolic blood pressure >90mmHg?

YES

1L 0.9% Sodium Chloride over 1 hour

! WARNING: high risk of severe hypokalaemia during treatment of DKA. Monitor regularly and follow pathway.

Potassium chloride may be required **if more than 1 litre** of Sodium Chloride has already been given to resuscitate a hypotensive patient - **See step 7**

NO

500ml 0.9% Sodium Chloride over 10-15min
THEN RE-CHECK BP

IF NOW >90 MOVE TO YES

IF STILL <90:

Further 500ml 0.9% Sodium Chloride over 10-15min
Request senior review and consider involving ICU

← Initial here to confirm patient has been started on correct fluid regime.

STEP 4

Start Fixed Rate IV Insulin Infusion

Prepare: Actrapid 50 units in 0.9% Sodium Chloride 49.5ml	Calculate the initial rate (0.1units/kg/hr)	Patients Weight:	kg	X 0.1 = Start rate	Units/hr	MAXIMUM 15 UNITS PER HOUR
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Continue at this rate until criteria for resolution of DKA in Step 9 have been met

Insulin prescription and administration chart

Date to start	Time to start	Insulin (route IV)	Infusion fluid	Prescriber's signature	Print name / Bleep number	Insulin batch no.	Actual start time & date	Pump asset No.	Infusion line changed (initial)	Signature	
										Adminis-tered by:	Checked by:
		Humulin S/ Actrapid 50 units Draw up in an Insulin syringe ONLY	Mixed in 49.5mL sodium chloride 0.9% (final volume – 50mL)								

☐ **Tick here if the patient normally takes long acting insulin.**
If the patient takes long acting insulin, e.g. Humulin I, Insulatard, Lantus, Levemir, this should be continued as normal.

! Prescribing advice: The long acting Insulin must be prescribed on dedicated insulin chart and must be written up on the patient's standard drug chart as a regular medication.

←Initial here to confirm that insulin has been prescribed and the intravenous infusion is running.

STEP 5	Further Assessment				
	- Carry out a full clinical examination. - Consider precipitating factors. Critical Care review – refer if any of the following are present: - blood ketones above 6mmol/L • venous bicarbonate level below 5mmol/L • venous or arterial pH below 7.1 - hypokalaemia on admission (below 3.5mmol/L) • GCS less than 12 • oxygen saturation below 92% on air (assuming normal baseline respiratory function) • systolic BP below 90 mmHg • Pulse over 100 or below 60 bpm.				Initial when complete →
	Choose relevant investigations (tick when requested):				
	ECG ☐	MSU ☐	CXR ☐	Blood Cultures ☐	
	☐ Refer to Diabetes Team				
	←Initial here to confirm that you have considered the necessary investigations and referred patient to the Diabetes Team.				
STEP 6	Fluid Replacement				
	Prescribe the following IV fluid regime alongside the insulin infusion for the next 8 hours. Be <u>cautious</u> if your patient is 18 – 25, elderly, pregnant or has heart failure or renal failure . Ensure prescription of 10% glucose to commence when glucose below 14.				
	1L 0.9% Sodium Chloride over 2 hours 1L 0.9% Sodium Chloride over 2 hours 1L 0.9% Sodium Chloride over 4 hours		! WARNING: high risk of severe hypokalaemia during treatment of DKA. Potassium must be added to each bag (unless K >5.5 mmol/l). See step 7 for further information.		
	←Initial here when infusion of the first bag has started and the fluid chart is completed.				
STEP 7	Further Monitoring				
	Continue to record values for blood glucose, blood ketones, <u>VENOUS</u> pH, potassium and rate of insulin per hour on the chart (see bottom of monitoring chart for suggested frequency).				
	RETURN TO THIS STEP REGULARLY AND MAKE THE FOLLOWING CHECKS				
	If BLOOD GLUCOSE falls below 14mmol/L		ADD 10% GLUCOSE at 125mL/h (Continue 0.9% Sodium Chloride at previous infusion rate unless there are concerns over fluid overload).		
	POTASSIUM Over 5.5 mmol/L Between 3.5–5.5 mmol/L Below 3.5 mmol/L		No added potassium. Every 1L of 0.9% sodium chloride to contain 20mmol potassium chloride. (infuse at a maximum rate of 10mmol KCl / hour) Seek immediate senior or critical care advice as may need additional potassium and admission to ITU.		
	If INCONTINENT or OLIGURIC (<0.5ml/kg/hr)		Consider urinary catheterisation.		
	If PERSISTANT VOMITING or REDUCED CONSCIOUSNESS		NG tube and airway protection.		
	If O₂ saturation is <92%		Do an ABG (as opposed to VBG).		
		←Initial here to confirm that the above has been noted, then move on to step 8.			
STEP 8	Assess Response to Treatment				
	1 hour after start of infusion				
	EVERY HOUR	Ketones should be falling by at least 0.5mmol/L per hr OR Bicarbonate should be rising by at least 3mmol/L per hr AND Blood glucose should be falling by at least 3mmol/L per hr		If this is not occurring, first check: • Cannula is patent • Insulin syringe driver is connected and working • Residual volume of insulin in the syringe driver is correct	
	If equipment is working but response to treatment remains inadequate, increase the insulin infusion rate by 1 unit/hour increments every hour until the target is achieved.				
	←Initial here to confirm that the above has been noted.				
!!	DON'T FORGET TO KEEP REFERRING BACK TO STEP 7 AND CHECKING THE FOLLOWING:				!
BLOOD GLUCOSE		POTASSIUM		URINE OUTPUT	
BLOOD GLUCOSE		POTASSIUM			
STEP 9	Target of Treatment: Resolution of Ketoacidosis				
	When Ketones <0.6mmol/L Enter value here →		When pH >7.3 and/or Bicarbonate >18mmol/L Enter value here →		
	DKA has resolved ONLY when both criteria are met.				
	Switch to a Variable Rate Insulin Infusion				
	When ketoacidosis has resolved, use the VRIII chart, starting with the standard regimen. Continue until the patient is eating and drinking.				
	←Initial here to confirm resolution of the DKA and commencement of Variable Rate Intravenous Insulin Infusion (VRIII).				

ADULT DIABETIC KETOACIDOSIS – HOURLY MONITORING CHART

TREATMENT AIMS:

1. Blood ketones to fall by at least 0.5mmol/L/h 2. Venous bicarbonate to rise by at least 3mmol/L/h 3. Blood glucose to fall by at least 3mmol/L/h

If this is NOT being achieved check insulin is being delivered correctly then INCREASE INSULIN RATE BY 1 unit/hour and review fluid requirement.

RESOLUTION OF DKA: Blood ketones <0.6mmol/L AND/OR venous pH >7.3 AND/OR venous bicarbonate ≥18mmol/L

Hours from start of treatment (h):	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Actual Time (HH:MM):	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:
	>20																								
Capillary Blood Glucose (mmol/L): (measure every hour and write value in appropriate box)	20																								
	19																								
	18																								
	17																								
	16																								
	15																								
	14																								
	13																								
IMPORTANT: ADD 10% GLUCOSE at 125mL /h WHEN CBG <14 (Continue 0.9% Sodium Chloride at previous infusion rate unless there are concerns over fluid overload.)	12																								
	11																								
	10																								
	9																								
	8																								
	7																								
	6																								
	5																								
TARGET GLUCOSE: 6 – 12 (If Glucose continues to fall despite addition of 10% Glucose then reduce insulin infusion rate as appropriate)	4																								
	3																								
	2																								
	1																								
Insulin rate (mL/h):																									
Blood ketones (mmol/L):																									
0.9% Sodium Chloride rate (mL/h):																									
10% Glucose rate (mL/h):																									
Urine output (mL/h):																									
VENOUS pH:																									
Venous potassium (mmol/L):																									
Venous Bicarbonate (mmol/L):																									

STEP 10

Conversion to Subcutaneous Insulin

When the patient is eating, drinking and metabolically stable, transfer to subcutaneous insulin. Ensure that this is started **before** IV insulin is discontinued.

☐

Tick here if the patient is not already established on insulin.
This is usually managed by the Specialist Diabetes Team. If the patient is newly diagnosed it is essential that they are seen by a member of the team prior to discharge.

☐

Tick here if the patient is already established on insulin.
The previous regimen should generally be restarted. The IV insulin infusion should continue until **30 minutes after the administration of subcutaneous insulin associated with a meal.**

Patients on basal bolus:

Long acting insulin analogues (if used) should be continued, and thus normal rapid acting insulin should be restarted at the next meal. IV insulin and fluids should be discontinued 30mins later. If basal insulin was stopped in error, contact the Diabetes Team for advice.

Patients on twice daily mixed insulin:

Reintroduce before breakfast or evening meal. Do not change at any other time. Maintain insulin infusion until 30 minutes after subcutaneous insulin given.

Patients on insulin pump (CSII):

Recommence at normal basal rate. Continue IV insulin until meal bolus given. Do not recommence CSII at bedtime.

←Initial here to confirm that an appropriate subcutaneous regime has been started.

END OF PATHWAY

Please file these pages in the patient notes as a record of treatment.

Diabetes Team Contact Details

York (772) 6091 / 5793

Scarborough (771) 2803 / 2804

For Diabetes use only

Date of admission		
Date seen by Diabetes Team		
Time seen by Diabetes Team		
Reason if > 24 hours before being seen (ie weekend, bank holiday, work load, not aware of admission)		
Sick day rules information provided	Yes ☐	No ☐
If no explain why:		
Follow up apt/Structured education	Yes ☐	No ☐
If no explain why:		
Ketone measuring equipment provided	Yes ☐	No ☐
Entry on EDN for GP's attention	Yes ☐	No ☐